

project proposal

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

library(dplyr)      # data manipulation
library(broom)      # tidy statistical output
library(corrplot)
```

Data

```

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

2. Cleaning

sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

Cleaning taxon_list

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

Cleaning water_quality

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
```

```

    diptera_chironomid,
    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

Focus Taxa

```

# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")

```

Figure 1: Invertebrate abundance as a function of DO

```
# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# creates median point location for each taxon
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    med_do_point = median(med_do, na.rm = TRUE),
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
  # uses DO_taxa data frame
  mapping = aes(x = med_do, # x-axis: med_do
    y = log_ave_lit, # y-axis: log_ave_lit
    color = taxon)) + # color by taxon

# first layer: points show individual abundance measurements
geom_point(shape = 16, # shape is circle
  size = 2, # size is 2
  alpha = 0.7) + # opacity is 0.7

# second layer: larger point shows the median location for each taxon
geom_point(data = DO_taxa_medians,
  mapping = aes(x = med_do_point,
```

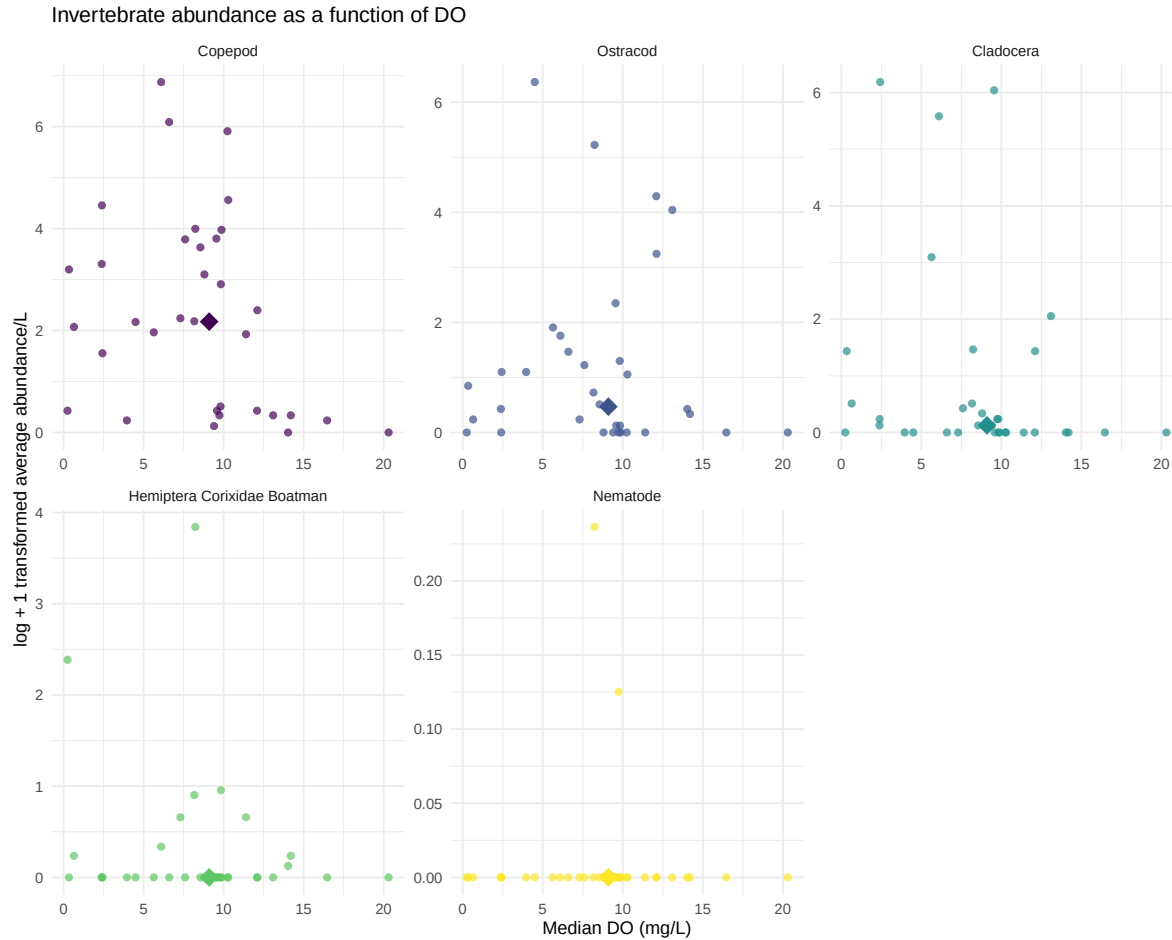
```

                                y = med_log_ave_lit),
    shape = 18,
    size = 5) +

# separate panels for each taxon with free x- and y-axis scales
facet_wrap(~ taxon, scales = "free") +
# use non-default colors for points
scale_color_viridis_d() +
labs(x = "Median DO (mg/L)", # relabel x-axis
     y = "log + 1 transformed average abundance/L", # relabel y-axis
     # creates descriptive title
     title = "Invertebrate abundance as a function of DO") +

# uses non-default theme
theme_minimal() +
# remove redundant legend
theme(legend.position = "none")
DO_taxa_plot

```



shows that individual taxa have noisy, weak, or inconsistent relationships with DO.

Figure 2: Variance of DO across sights

```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <- aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)

# base layer
```

```

DO <- ggplot(data = aq_ins_clean_fig2,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_do)) +
# first layer: boxplot
geom_boxplot(width = 0.35,
  outlier.shape = NA) +
# second layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.12,
  shape = 21,
  alpha = 0.6) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median DO (mh/L)",
  title = "Site DO levels") +
# different theme
theme_bw()
# show plot
DO

```

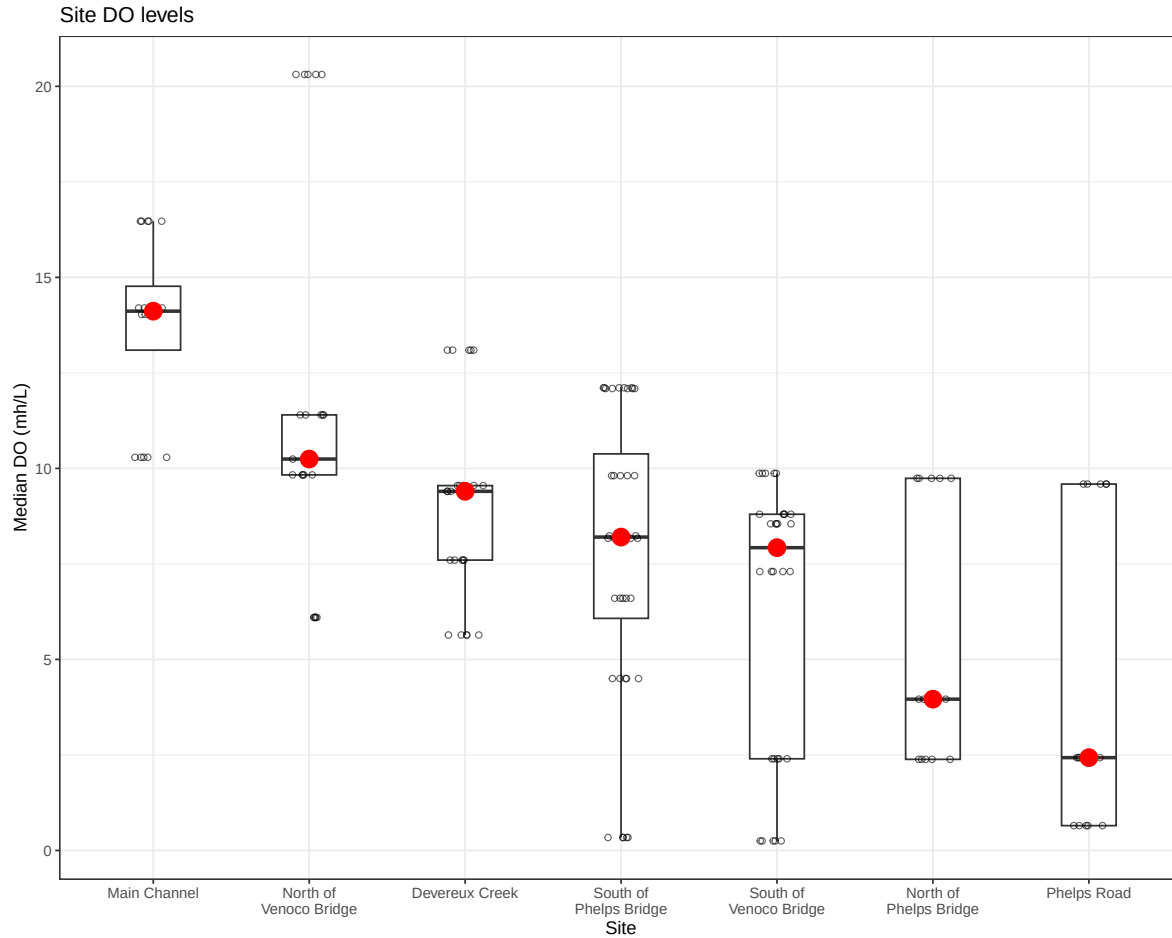



Figure 3: Aquatic Invert. Abundance Across Sites

```
# creating new object from clean aquatic inverts
abundance_by_site <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include taxa of interest
  filter(taxon %in% focus_taxa)
abundance <- ggplot(data = abundance_by_site,
  # x-axis: site
```

```

# y-axis: mean abundance/L
# color by taxon
mapping = aes(x = site_full,
              y = log_ave_lit,
              color = taxon)) +
# first layer: jitter points to represent observations
geom_jitter(width = 0.15,
            height = 0,
            alpha = 0.7) +
# add median point for each site within each taxon
stat_summary(fun = median,
            geom = "point",
            shape = 18,
            size = 3,
            color = "black") +
# facet by taxon
facet_wrap(~ taxon) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# non-default colors
scale_color_viridis_d() +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Aquatic Invert. Abundance Across Sites") +
theme_bw() +
# remove legend
theme(legend.position = 'none',
      # fix x-axis for readability
      axis.text.x = element_text(angle = 45, hjust = 1))

```

```

# base layer

```

```

abundance <- ggplot(data = abundance_by_site,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   # color by taxon
                   mapping = aes(x = site_full,
                                y = log_ave_lit,
                                color = taxon)) +
# first layer: jitter points to represent observations
geom_jitter(width = 0.15,

```

```

        height = 0,
        alpha = 0.7) +
# facet by taxon
facet_wrap(~ taxon) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# non-default colors
scale_color_viridis_d() +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Aquatic Invert. Abundance Across Sites") +
theme_bw() +
# remove legend
theme(legend.position = 'none',
      # fix x-axis for readability
      axis.text.x = element_text(angle = 45, hjust = 1))
abundance

```

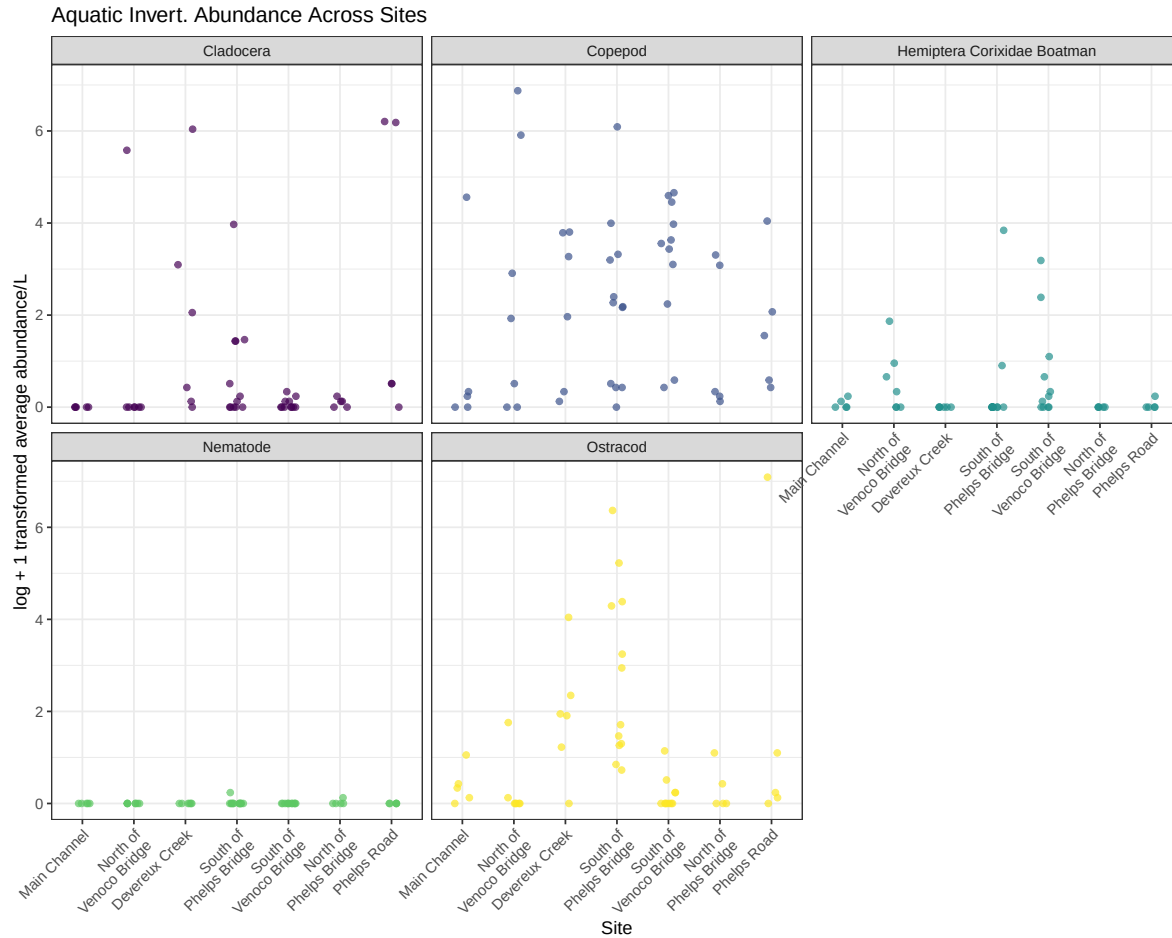


Figure 4: Abundance over time

```
# create object for plotting total abundance over time
abundance_time <- aq_ins_clean |>
  # group by sampling date and taxon
  group_by(date, taxon) |>
  # total abundance across all sites for each date and taxon
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create log-transformed total abundance column
  mutate(log_total_ave_lit = log(total_ave_lit + 1)) |>
  # make taxon names easier to read
```

```

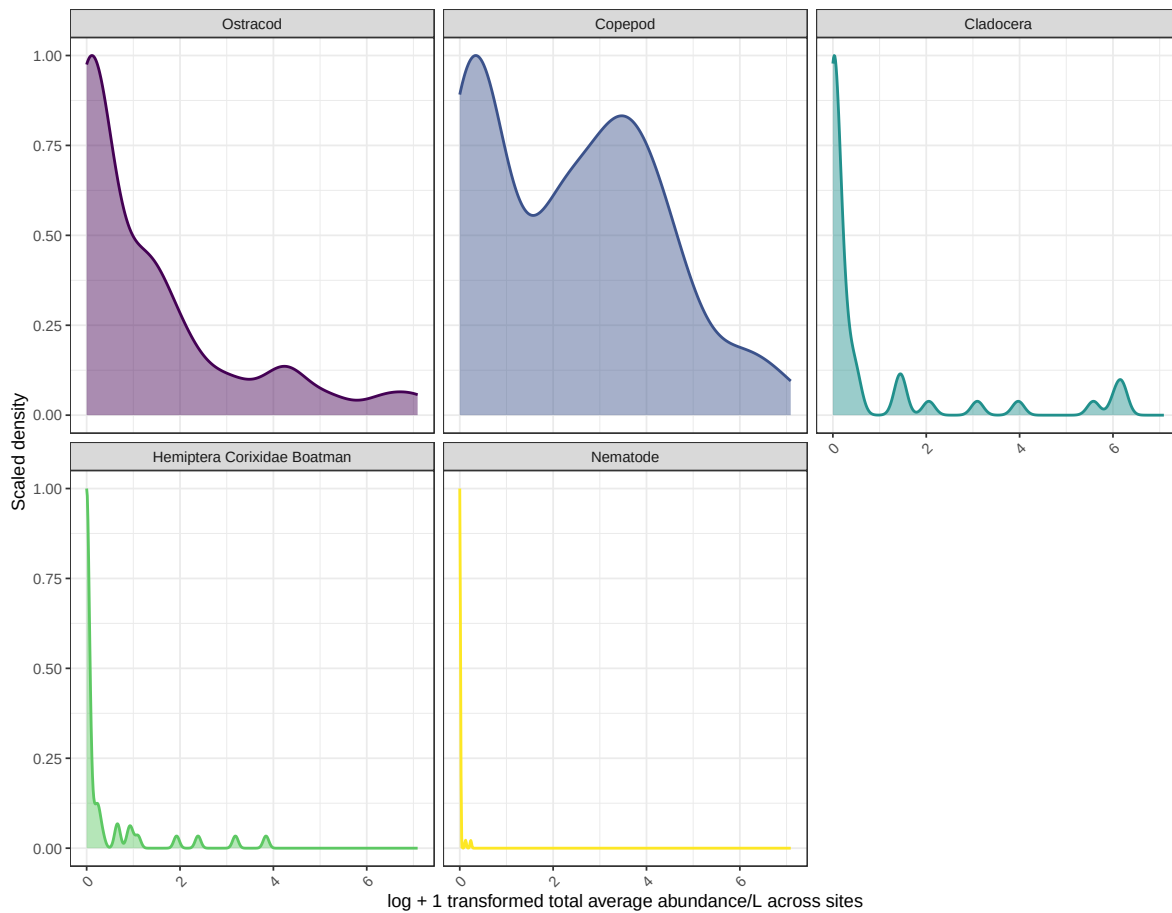
mutate(taxon = str_replace_all(taxon, "_", " "),
       taxon = str_to_title(taxon)) |>
# order taxa by maximum total abundance
mutate(taxon = fct_reorder(taxon, total_ave_lit, .fun = max, .desc = TRUE)) |>
# include main taxon groups of focus
filter(taxon %in% focus_taxa)

# base layer
abundance_time_plot <- ggplot(data = abundance_time,
                             # x-axis: log_total_ave_lit
                             # fill and color by taxon
                             mapping = aes(x = log_total_ave_lit,
                                             fill = taxon,
                                             color = taxon)) +
# scaled density makes y-axis consistent across facets
geom_density(aes(y = after_stat(scaled)),
             alpha = 0.45,
             linewidth = 0.8,
             adjust = 0.8) +
facet_wrap(~ taxon) + # facet by taxon
scale_fill_viridis_d() + # non-default colors
scale_color_viridis_d() + # non-default colors
labs(x = "log + 1 transformed total average abundance/L across sites",
     y = "Scaled density",
     title = "Distribution of total invertebrate abundance over time") +
# use clean theme
theme_bw() +
# remove redundant legend
theme(legend.position = "none",
      axis.text.x = element_text(angle = 45,
                                  hjust = 1))

# show plot
abundance_time_plot

```

Distribution of total invertebrate abundance over time



correlation analysis

```
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  select(date, site, taxon, log_ave_lit, med_do) |>
  pivot_wider(
    names_from = taxon,
    values_from = log_ave_lit
  )

# calculate correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  select(-date, -site) |>
```

```

summarize(across(-med_do, ~ cor(.x, med_do, use = "complete.obs"))) |>
pivot_longer(
  cols = everything(),
  names_to = "taxon",
  values_to = "correlation"
)

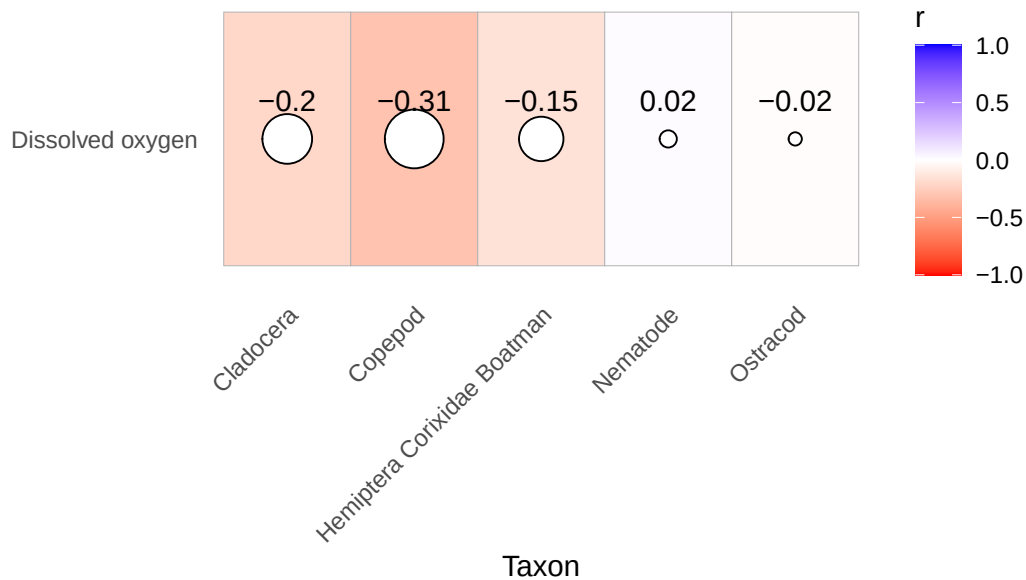
ggplot(cor_do, aes(x = taxon, y = "Dissolved oxygen", fill = correlation)) +
  geom_tile(color = "grey70") +
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +

  # move labels ABOVE circles
  geom_text(aes(label = round(correlation, 2)),
            nudge_y = 0.15, # <-- this moves text up
            size = 4) +

  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-1, 1),
    name = "r"
  ) +
  scale_size(range = c(2, 10), guide = "none") +
  labs(
    x = "Taxon",
    y = NULL,
    title = "Correlation between dissolved oxygen and invertebrate abundance"
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

```

Correlation between dissolved oxygen and invertebrate



Summed abundance across focus taxa vs DO

```
focus_taxa <- c("Cladocera", "Copepod",
                 "Hemiptera Corixidae Boatman",
                 "Nematode", "Ostracod")

total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    log_total_abundance = log10(total_abundance + 1),
    .groups = "drop"
  )

ggplot(total_abundance,
       aes(x = med_do,
           y = log_total_abundance)) +

  geom_point(color = "magenta3",
            size = 3,
```



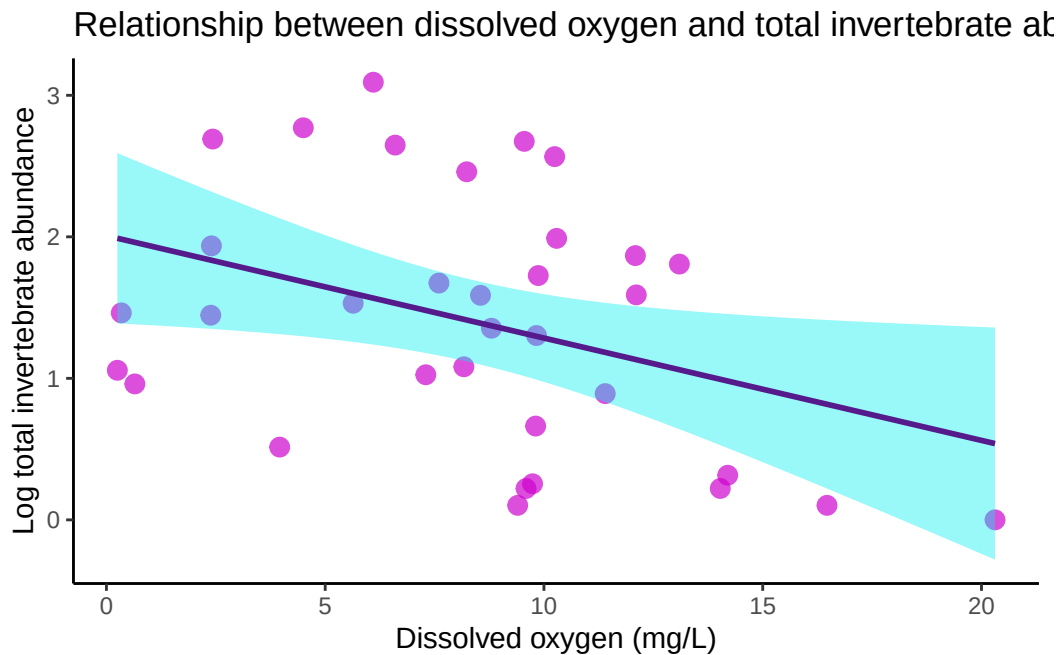
```

    alpha = 0.7) +
  geom_smooth(method = "lm",
             se = TRUE,
             color = "purple4",
             fill = "cyan2") +

  labs(
    x = "Dissolved oxygen (mg/L)",
    y = "Log total invertebrate abundance",
    title = "Relationship between dissolved oxygen and total invertebrate abundance"
  ) +

  theme_classic()

```



When taxa are combined, a clearer negative relationship emerges.

```

cor.test(
  total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "pearson"
)

```

Pearson's product-moment correlation

```
data: total_abundance$log_total_abundance and total_abundance$med_do  
t = -2.2882, df = 32, p-value = 0.02887  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
-0.63289820 -0.04217141  
sample estimates:  
cor  
-0.3749894
```

Binned DO levels

```
total_abundance <- total_abundance |>  
  mutate(  
    do_category = case_when(  
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",  
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",  
      TRUE ~ "High DO"  
    )  
  )  
total_abundance$do_category <- factor(  
  total_abundance$do_category,  
  levels = c("Low DO", "Medium DO", "High DO")  
)  
  
ggplot(total_abundance,  
  aes(x = do_category,  
      y = log_total_abundance,  
      fill = do_category)) +  
  
  geom_boxplot(alpha = 0.7,  
    outlier.color = "black",  
    outlier.size = 2) +  
  
  # optional: add jittered points to show raw data  
  geom_jitter(width = 0.15,  
    alpha = 0.5,  
    size = 2,  
    color = "black") +
```

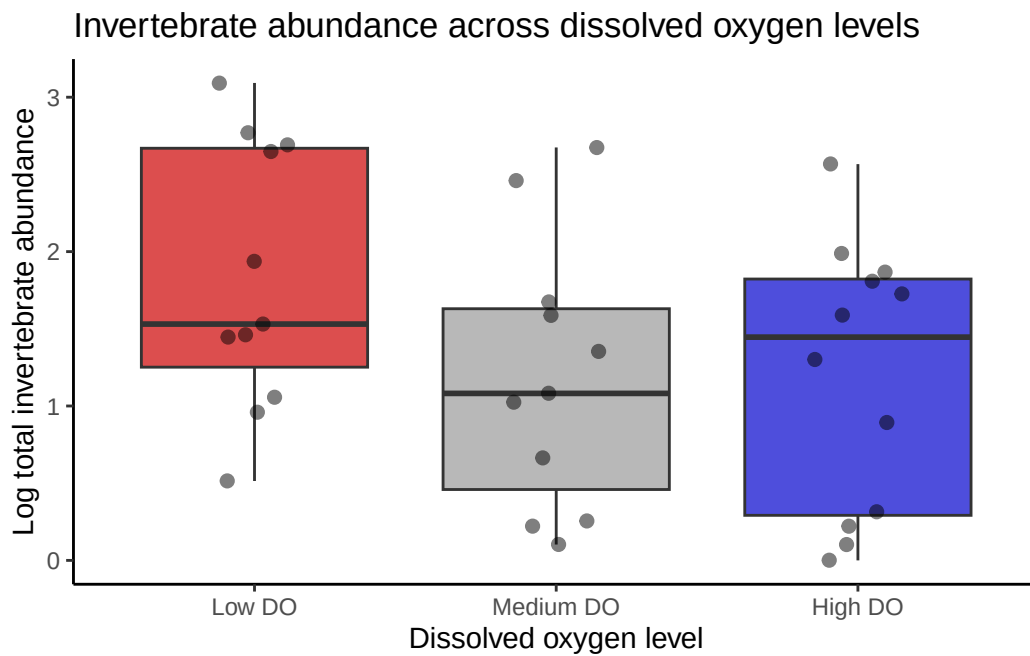
```

labs(
  x = "Dissolved oxygen level",
  y = "Log total invertebrate abundance",
  title = "Invertebrate abundance across dissolved oxygen levels"
) +

scale_fill_manual(values = c(
  "Low DO" = "red3",
  "Medium DO" = "grey60",
  "High DO" = "blue3"
)) +

theme_classic() +
theme(legend.position = "none")

```



Community abundance varies continuously rather than discretely across DO levels.

```

# Pearson correlation for continuous DO and total community abundance
cor.test(total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "pearson")

```

Pearson's product-moment correlation

```
data: total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
      cor
-0.3749894
```

```
# Kruskal-Wallis test comparing abundance across DO categories
kruskal.test(log_total_abundance ~ do_category,
             data = total_abundance)
```

Kruskal-Wallis rank sum test

```
data: log_total_abundance by do_category
Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
```

```
composition_plot <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),

    # reorder factor levels
    do_category = factor(do_category,
                        levels = c("Low DO",
                                   "Medium DO",
                                   "High DO"))
  ) |>
  group_by(do_category, taxon) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = do_category,
```

```

    y = total_abundance,
    fill = taxon)) +

geom_col(position = "fill") +

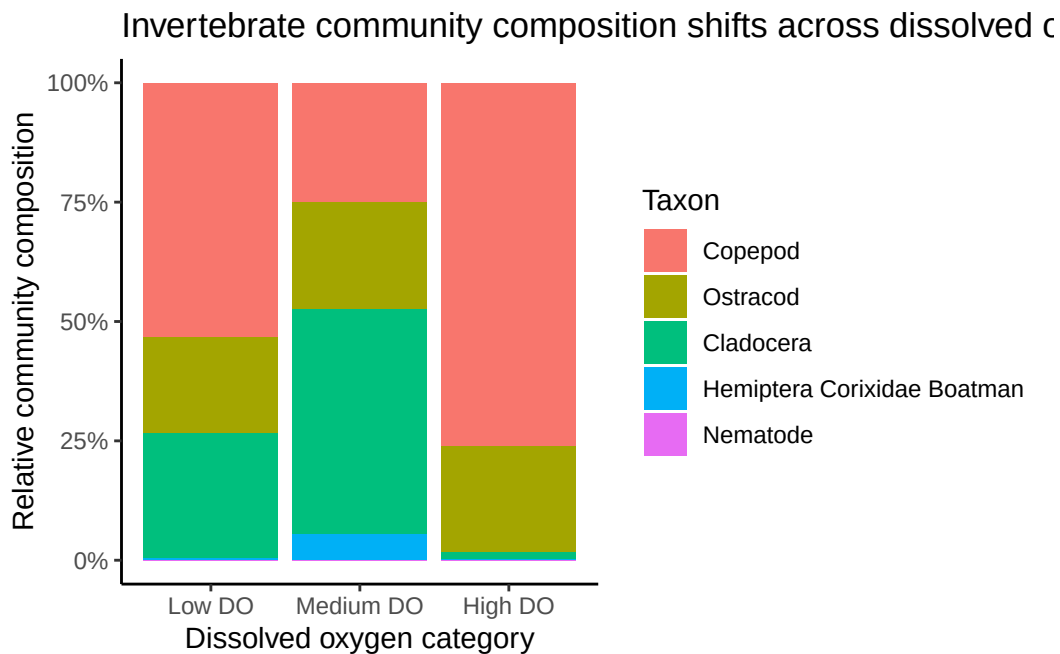
scale_y_continuous(labels = scales::percent_format()) +

labs(
  x = "Dissolved oxygen category",
  y = "Relative community composition",
  fill = "Taxon",
  title = "Invertebrate community composition shifts across dissolved oxygen levels"
) +

theme_classic()

composition_plot

```



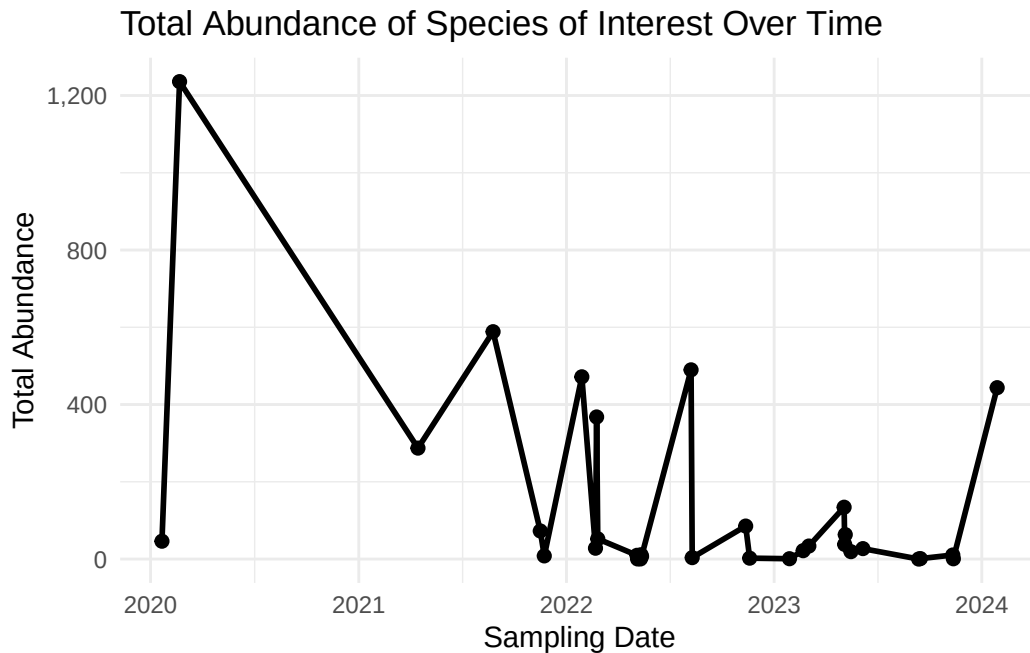
Medium DO sites show a more even distribution of taxa and greater relative abundance of Cladocera and Ostracods, while low and high DO conditions are dominated primarily by Copepods.

Time-series plot (all species)

```
# species/taxa to include in the total abundance figure
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

# create total abundance over time
total_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% species_interest) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date only, so abundance is summed across sites and taxa
  group_by(date) |>
  # sums average abundance per liter across all taxa of interest
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# plot total abundance over time
ggplot(total_abundance_time, aes(x = date, y = total_abundance)) +
  # adds a line connecting total abundance values over time
  geom_line(linewidth = 1) +
  # adds points for each sampling date
  geom_point(size = 2) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Total Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Total Abundance"
  ) +
  # uses a clean theme
  theme_minimal()
```



Time-series area plot

```
# species/taxa to include in the area plot
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

# create abundance over time by taxon
taxon_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% species_interest) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date and taxon so each species stays separate
  group_by(date, taxon) |>
  # sums across sites, but not across taxa
  summarise(
```

```

    ave_lit = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# create time series area plot by taxon
ggplot(taxon_abundance_time, aes(x = date, y = ave_lit, fill = taxon)) +
  # creates stacked area plot with a different color for each taxon
  geom_area(alpha = 0.8) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Average Abundance per Liter",
    fill = "Taxon"
  ) +
  # uses a clean theme
  theme_minimal()

```

